

# FASTCash 2.0 APT38 / BeagleBoyz

## Mapping a Real Financial APT Campaign to the MITRE ATT&CK; Framework

**CASE SELECTION** APT38 is a North Korean state-sponsored threat group specializing in financial cyber operations. MITRE lists BeagleBoyz as an associated group name and notes that APT38 has targeted banks, SWIFT endpoints and ATMs worldwide.

**Campaign / advisory focus:** FASTCash 2.0 - North Korea's BeagleBoyz Robbing Banks (CISA AA20-239A, August 26, 2020).

**Industry scope:** Banking and financial services. This makes the case directly relevant to a policy-governance project positioned for the banking sector, where payment controls, access procedures, incident response and transaction integrity are highly regulated.

**Framework:** MITRE ATT&CK; Enterprise. The mapping below uses behaviors that MITRE attributes to APT38 with CISA's FASTCash 2.0 advisory as the cited source, rather than treating every behavior ever associated with APT38 as part of this one case.

# Why FASTCash 2.0 is a strong banking case

FASTCash operations are financially motivated intrusions associated with North Korean state-sponsored actors. The BeagleBoyz activity cluster is associated by MITRE with APT38. The campaign is especially useful for ATT&CK; mapping because it demonstrates behaviors across several stages of an intrusion: gaining access, executing commands, establishing persistence, discovering the environment, collecting information, moving through systems, reducing forensic evidence and affecting financial operations.

**IMPORTANT DISTINCTION** APT38 is the threat group. FASTCash 2.0 is the selected real-world banking operation/advisory used as the case study. MITRE ATT&CK; is the framework used to classify the adversary's observed behaviors.

## ATT&CK; in plain language

**Tactic** = the attacker's objective or 'why.' **Technique** = the method or 'how.' **Sub-technique** = a more specific form of that method. **Procedure** = the real-world example of how this actor actually used the technique.

Example: if an attacker sends a malicious attachment to get into a bank, the objective is **Initial Access**; the technique family is **Phishing**; and the specific sub-technique is **Spearphishing Attachment (T1566.001)**.

## Observed behaviors mapped to ATT&CK;

The following mapping separates the ATT&CK; classification from the procedure evidence. The first nine rows are behaviors MITRE attributes to APT38 with CISA FASTCash 2.0 as a supporting reference on the APT38 group page. The three SWIFT data-manipulation rows are included because they are highly relevant to APT38's banking tradecraft, but MITRE cites separate APT38 research for those specific procedures; they should be read as broader APT38 financial-operation context rather than as claims unique to the CISA advisory.

| Tactic            | ID        | Technique                                        | Procedure / evidence                                                                                                               |
|-------------------|-----------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Initial Access    | T1566.001 | Phishing: Spearphishing Attachment               | APT38 has conducted spearphishing campaigns using malicious email attachments.                                                     |
| Execution         | T1059.001 | Command and Scripting Interpreter: PowerShell    | APT38 has used PowerShell to execute commands and perform operational tasks.                                                       |
| Persistence       | T1543.003 | Create or Modify System Process: Windows Service | APT38 has installed a new Windows service to establish persistence.                                                                |
| Credential Access | T1110     | Brute Force                                      | APT38 has used brute-force techniques when passwords or password hashes were unavailable.                                          |
| Discovery         | T1083     | File and Directory Discovery                     | APT38 has enumerated files/directories and searched specific locations on compromised hosts.                                       |
| Discovery         | T1135     | Network Share Discovery                          | APT38 has enumerated network shares on compromised hosts.                                                                          |
| Collection        | T1005     | Data from Local System                           | APT38 has collected data from compromised hosts.                                                                                   |
| Defense Evasion   | T1070.004 | Indicator Removal: File Deletion                 | APT38 has removed malware, tools and other non-native files to reduce its footprint after intrusion activity.                      |
| Defense Evasion   | T1070.006 | Indicator Removal: Timestamp                     | APT38 has modified timestamps to resemble other files in the same folder.                                                          |
| Impact            | T1565.001 | Data Manipulation: Stored Data Manipulation      | MITRE documents APT38 using DYEPACK to create, delete and alter database records used for SWIFT transactions.                      |
| Impact            | T1565.002 | Data Manipulation: Transmitted Data Manipulation | APT38 has used DYEPACK to manipulate SWIFT messages while they were being transmitted to a printer.                                |
| Impact            | T1565.003 | Data Manipulation: Runtime Data Manipulation     | APT38 used DYEPACK.FOX to alter PDF data as it was accessed, removing traces of fraudulent SWIFT transactions from what users saw. |

## Reading the map as an intrusion sequence

- 1. Gain a foothold.** Spearphishing attachments provide one documented initial-access path associated with APT38.
- 2. Execute activity.** PowerShell provides a built-in command environment that can be abused for attacker operations.
- 3. Maintain access.** A Windows service can provide persistence on a compromised system.
- 4. Expand access and knowledge.** Brute force, file/directory discovery and network-share discovery help the actor locate accounts, systems and resources.
- 5. Collect useful information.** Data from local systems can support the adversary's understanding of the banking environment.
- 6. Hide evidence.** File deletion and timestamp manipulation reduce visible traces and complicate forensic analysis.
- 7. Affect financial integrity.** APT38's broader banking tradecraft includes manipulation of stored, transmitted and runtime SWIFT-related data, illustrating why transaction integrity is a critical impact concern.

**ANALYST INTERPRETATION** ATT&CK; is not a rigid timeline: attackers can repeat, skip or combine techniques. The value of the framework is that it gives defenders a common vocabulary for describing observed behavior and identifying where detection and controls should be placed.

### Why this matters specifically to banking

Financial institutions depend on trusted identities, payment systems, transaction records, network segmentation and reliable audit trails. APT38's tradecraft demonstrates risk to each of these areas. A successful intrusion can therefore become more than an IT problem: it can threaten the integrity of payment operations, financial records and evidence used by fraud, compliance and incident-response teams.

## What a bank should learn from the mapping

The mapping can be converted into defensive priorities. Phishing-resistant authentication and email controls reduce initial-access risk. PowerShell logging and endpoint monitoring improve execution visibility. Service creation monitoring can expose persistence. Authentication analytics can identify brute-force activity. File, share and host discovery telemetry can reveal reconnaissance. Finally, immutable logging, transaction reconciliation and separation of duties are especially important where an attacker may try to alter or conceal payment data.

| ATT&CK; area                   | Banking control emphasis                                                                                |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Initial Access                 | Email security, attachment inspection, user awareness, phishing-resistant MFA.                          |
| Execution / Persistence        | Endpoint detection, PowerShell telemetry, application control, service-creation alerts.                 |
| Credential Access              | Strong authentication, lockout/rate controls, privileged-access management, anomalous-login detection.  |
| Discovery / Collection         | Network segmentation, least privilege, monitoring of unusual share/file enumeration.                    |
| Defense Evasion                | Centralized/immutable logs, EDR tamper protection, alerts for suspicious deletion or timestamp changes. |
| Impact / transaction integrity | Dual control, SWIFT/payment monitoring, independent reconciliation, protected audit trails.             |

**CONNECTION TO POLICY GOVERNANCE** For a banking-focused policy assistant, this case reinforces why employees need fast access to current procedures for phishing reporting, privileged access, payment verification, incident escalation, evidence preservation and suspicious transaction handling.

## Assignment-ready conclusion

APT38 / BeagleBoyz is an appropriate case for a banking-sector MITRE ATT&CK; mapping because the group is explicitly associated with financial cyber operations. FASTCash 2.0 provides a real-world context for understanding how adversary behaviors can be organized using ATT&CK; tactics and techniques. The mapping shows that a financial intrusion is not one isolated action: it can involve initial access, command execution, persistence, credential attacks, discovery, collection, defense evasion and ultimately actions that threaten the integrity of financial operations.

The key value of MITRE ATT&CK; is that it converts a complex incident into a structured behavioral model. Security teams can use that model to communicate what an adversary did, connect evidence across systems, identify detection gaps and prioritize controls.

### SOURCE NOTES

1. **CISA.** AA20-239A, *FASTCash 2.0: North Korea's BeagleBoyz Robbing Banks*, published August 26, 2020. <https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-239a>

2. **MITRE ATT&CK.**; APT38 (G0082). MITRE identifies APT38 as a North Korean state-sponsored group specializing in financial cyber operations, lists BeagleBoyz as an associated group, and provides the ATT&CK; technique/procedure mappings used in this report. <https://attack.mitre.org/groups/G0082/>

3. **MITRE ATT&CK.**; T1565.001 Stored Data Manipulation; T1565.002 Transmitted Data Manipulation; T1565.003 Runtime Data Manipulation. These pages document APT38's SWIFT-related data-manipulation procedure examples.

**METHOD NOTE** This report intentionally distinguishes campaign-supported behaviors from broader APT38 tradecraft. It does not assume that every technique on MITRE's complete APT38 profile occurred in FASTCash 2.0.